MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR FUEL UNLOADING STATION

DESIGN CALCULATION OF FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-1627

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	26-10-2023	AS BUILT	SHD	SGC	PRI	
C01	29-06-2020	ISSUED FOR CONSTRUCTION	SHD	SGC	PRI	
A01	29-05-2020	FIRST ISSUE	SHD	SGC	PRI	

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Ma'aden Document Number:

MD-216-49PR-EG-CV-CAL-1627 D01

Outotec Document Number:

900707 GAU01 WG01 CED01 45010 OU100682336 2

L&T Document Number:

O19045 008 C K 24291 003 2

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	29-05-2020	29-06-2020	26-10-2023				Date	29-05-2020	29-06-2020	26-10-2023		
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Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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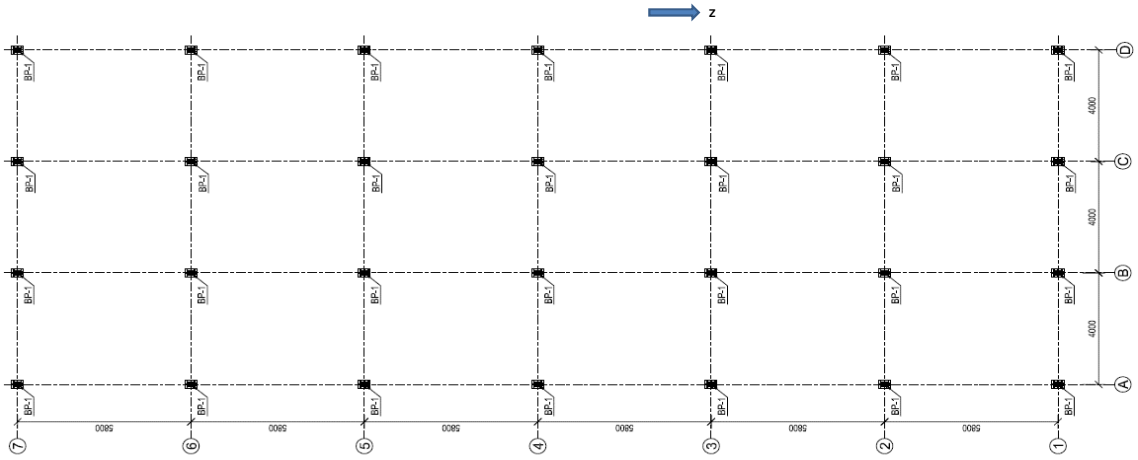
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1. INTENT

The document is prepared for Design Calculation of Column Foundation for **FUEL UNLOADING STATION**

2. SCOPE

The document is prepared for Design Calculation of Column Foundation for FUEL UNLOADING STATION
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The foundation is designed for wind load as seismic load is less than wind load.(Refer MD-216-49PR-EG-ST-CAL-2418- for Fuel unloading station design calculation)
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018
Net Safe Bearing Capacilty is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered



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3. Design of FUEL UNLOADING STATION FOUNDATION

3.1 Design of Foundation mkd. F1 (GRID B & C 2,3,4,5,6)

Load Detail	C (2,3,4,5,6)		
	FX (T)	FY (T)	FZ (T)
1 DL	0.0	4.0	0.0
2 LLROOF	0.0	3.6	0.0
21 WL1(+X)CASE-A	2.2	3.7	0.0
22 WL2(+X)CASE-B	2.2	0.6	0.0
23 WL3(-X)CASE-A	-2.2	-1.0	0.0
24 WL4(-X)CASE-B	-2.2	-4.0	0.0
25 WL5(+Z)CASE-A	-0.1	-5.7	0.8
26 WL6(+Z)CASE-B	-0.1	-6.0	0.8
27 WL7(-Z)CASE-A	-0.1	-5.7	-0.8
28 WL8(-Z)CASE-B	-0.1	-6.0	-0.8

STRENGTH DESIGN	C (2,3,4,5,6)		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4 DL	0.00	5.55	0.00
1.4 DL + 1.7 LL	0.1	11.7	0.0
1.2 DL + 1.6 WL1(+)x + LL	3.6	14.2	0.0
1.2 DL + 1.6 WL2(+)x + LL	3.5	6.9	0.0
1.2 DL + 1.6 WL3(-)x + LL	-3.5	6.8	0.0
1.2 DL + 1.6 WL4(-)x + LL	-3.4	1.9	0.0
1.2 DL + 1.6 WL5(+)z + LL	0.0	-0.7	1.3
1.2 DL + 1.6 WL6(+)z + LL	-0.1	-1.3	1.3
1.2 DL + 1.6 WL7(-)z + LL	0.0	-0.7	-1.3
1.2 DL + 1.6 WL8(-)z + LL	-0.1	-1.3	-1.3
0.9 DL + 1.6 WL1(+)x	3.6	9.4	0.0
0.9 DL + 1.6 WL2(+)x	3.5	4.5	0.0
0.9 DL + 1.6 WL3(-)x	-3.5	2.0	0.0
0.9 DL + 1.6 WL4(-)x	-3.5	-2.9	0.0
0.9 DL + 1.6 WL5(+)z	-0.1	-5.5	1.3
0.9 DL + 1.6 WL6(+)z	-0.1	-6.1	1.3
0.9 DL + 1.6 WL7(-)z	-0.1	-5.5	-1.3
0.9 DL + 1.6 WL8(-)z	-0.1	-6.1	-1.3

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DESIGN OF FOUNDATION MKD. F1

	About-X	About-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.85m	1.85m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

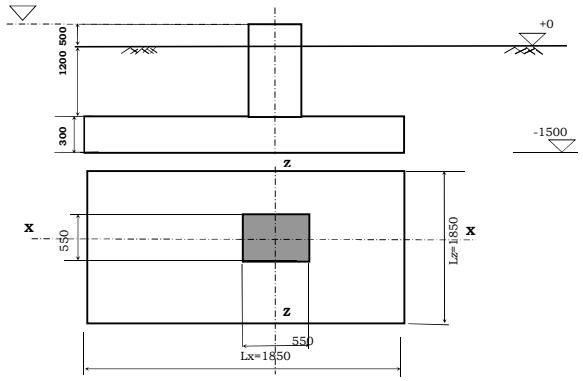
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.85*1.85*0.3*2.5)+(0.55*0.55*1.7*2.5) = 3.85T

Weight of soil = ((1.85*1.85)-(0.55*0.55))*1.2*1.8 = 6.74T

U/S Base Pl Total = 10.59T



Design Input:

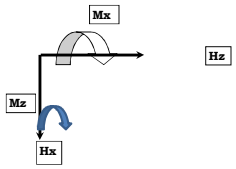
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = ######## 0.217m

Effective depth of slab along Z direction = ######## 0.201m

Clear projection along Z direction = 0.65m

Clear projection along X direction = 0.65m



WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.04	7.59	0.00	0.0	0.00	18.18	0.04	0.00	0.00	0.08
DL+0.75*LL+0.75*WL1(+)x	1.70	9.43	0.00	0.0	0.00	20.02	1.70	0.00	0.00	3.40
DL+0.75*LL+0.75*WL2(+)x	1.66	7.11	0.00	0.0	0.00	17.70	1.66	0.00	0.00	3.31
DL+0.75*LL+0.75*WL3(-)x	-1.62	5.93	0.00	0.0	0.00	16.52	-1.62	0.00	0.00	3.23
DL+0.75*LL+0.75*WL4(-)x	-1.60	3.66	0.00	0.0	0.00	14.25	-1.60	0.00	0.00	3.21
DL+0.75*LL+0.75*WL5(+)z	-0.01	2.42	0.60	0.0	0.00	13.01	-0.01	1.19	0.60	0.02
DL+0.75*LL+0.75*WL6(+)z	-0.01	2.15	0.60	0.0	0.00	12.74	-0.01	1.19	0.60	0.03
DL+0.75*LL+0.75*WL7(-)z	-0.01	2.42	-0.60	0.0	0.00	13.01	-0.01	1.19	-0.60	0.02
DL+0.75*LL+0.75*WL8(-)z	-0.01	2.15	-0.60	0.0	0.00	12.74	-0.01	1.19	-0.60	0.03
0.6*DL+WL1(+)x	2.22	6.04	0.00	0.0	0.00	16.63	2.22	0.00	0.00	4.44
0.6*DL+WL2(+)x	2.16	2.95	0.00	0.0	0.00	13.54	2.16	0.00	0.00	4.33
0.6*DL+WL3(-)x	-2.20	1.38	0.00	0.0	0.00	11.97	-2.20	0.00	0.00	4.40
0.6*DL+WL4(-)x	-2.18	-1.65	0.00	0.0	0.00	8.94	-2.18	0.00	0.00	4.36
0.6*DL+WL5(+)z	-0.06	-3.31	0.79	0.0	0.00	7.28	-0.06	1.59	0.79	0.11
0.6*DL+WL6(+)z	-0.06	-3.67	0.79	0.0	0.00	6.92	-0.06	1.59	0.79	0.12
0.6*DL+WL7(-)z	-0.06	-3.31	-0.79	0.0	0.00	7.28	-0.06	1.59	-0.79	0.11
0.6*DL+WL8(-)z	-0.06	-3.67	-0.79	0.0	0.00	6.92	-0.06	1.59	-0.79	0.12

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Overturning & Sliding Check (Coeffecient, μ = 0.4)			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+LL	18.18	0.04	0.08	0.00	0.00	200.20	NA	173.14	NA
DL+0.75*LL+0.75*WL1(+)x	20.02	1.70	3.40	0.00	0.00	5.45	NA	4.72	NA
DL+0.75*LL+0.75*WL2(+)x	17.70	1.66	3.31	0.00	0.00	4.95	NA	4.28	NA
DL+0.75*LL+0.75*WL3(-)x	16.52	-1.62	3.23	0.00	0.00	4.73	NA	4.09	NA
DL+0.75*LL+0.75*WL4(-)x	14.25	-1.60	3.21	0.00	0.00	4.11	NA	3.56	NA
DL+0.75*LL+0.75*WL5(+)z	13.01	-0.01	0.02	0.60	1.19	553.31	10.11	478.54	8.74
DL+0.75*LL+0.75*WL6(+)z	12.74	-0.01	0.03	0.60	1.19	448.86	9.90	388.21	8.56
DL+0.75*LL+0.75*WL7(-)z	13.01	-0.01	0.02	-0.60	1.19	553.31	10.11	478.54	8.74
DL+0.75*LL+0.75*WL8(-)z	12.74	-0.01	0.03	-0.60	1.19	448.86	9.90	388.21	8.56
0.6*DL+WL1(+)x	16.63	2.22	4.44	0.00	0.00	3.46	NA	2.99	NA
0.6*DL+WL2(+)x	13.54	2.16	4.33	0.00	0.00	2.89	NA	2.50	NA
0.6*DL+WL3(-)x	11.97	-2.20	4.40	0.00	0.00	2.52	NA	2.18	NA
0.6*DL+WL4(-)x	8.94	-2.18	4.36	0.00	0.00	1.90	NA	1.64	NA
0.6*DL+WL5(+)z	7.28	-0.06	0.11	0.79	1.59	58.80	4.25	50.85	3.67
0.6*DL+WL6(+)z	6.92	-0.06	0.12	0.79	1.59	53.09	4.03	45.91	3.49
0.6*DL+WL7(-)z	7.28	-0.06	0.11	-0.79	1.59	58.80	4.25	50.85	3.67
0.6*DL+WL8(-)z	6.92	-0.06	0.12	-0.79	1.59	53.09	4.03	45.91	3.49

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ULTIMATE STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4 DL	0.00	5.55	0.00	0.0	0.00	16.142	0.00	0.01
1.4 DL + 1.7 LL	0.07	11.71	0.00	0.0	0.00	22.303	0.00	0.14
1.2 DL + 1.6 WL1(+)x + LL	3.60	14.24	0.00	0.0	0.00	24.829	0.00	7.20
1.2 DL + 1.6 WL2(+)x + LL	3.51	6.89	0.00	0.0	0.00	17.485	0.00	7.01
1.2 DL + 1.6 WL3(-)x + LL	-3.47	6.78	0.00	0.0	0.00	17.375	0.00	6.95
1.2 DL + 1.6 WL4(-)x + LL	-3.44	1.93	0.00	0.0	0.00	12.524	0.00	6.89
1.2 DL + 1.6 WL5(+)z + LL	-0.05	-0.71	1.27	0.0	0.00	9.877	2.54	0.09
1.2 DL + 1.6 WL6(+)z + LL	-0.05	-1.29	1.27	0.0	0.00	9.296	2.54	0.10
1.2 DL + 1.6 WL7(-)z + LL	-0.05	-0.71	-1.27	0.0	0.00	9.877	2.54	0.09
1.2 DL + 1.6 WL8(-)z + LL	-0.05	-1.29	-1.27	0.0	0.00	9.296	2.54	0.10
0.9 DL + 1.6 WL1(+)x	3.55	9.43	0.00	0.0	0.00	20.015	0.00	7.11
0.9 DL + 1.6 WL2(+)x	3.46	4.48	0.00	0.0	0.00	15.069	0.00	6.93
0.9 DL + 1.6 WL3(-)x	-3.52	1.97	0.00	0.0	0.00	12.561	0.00	7.04
0.9 DL + 1.6 WL4(-)x	-3.49	-2.88	0.00	0.0	0.00	7.711	0.00	6.98
0.9 DL + 1.6 WL5(+)z	-0.09	-5.53	1.27	0.0	0.00	5.063	2.54	0.18
0.9 DL + 1.6 WL6(+)z	-0.10	-6.11	1.27	0.0	0.00	4.483	2.54	0.19
0.9 DL + 1.6 WL7(-)z	-0.09	-5.53	-1.27	0.0	0.00	5.063	2.54	0.18
0.9 DL + 1.6 WL8(-)z	-0.10	-6.11	-1.27	0.0	0.00	4.483	2.54	0.19

Check for Bearing Pressure

Load Combination & Calculation of Base pressure

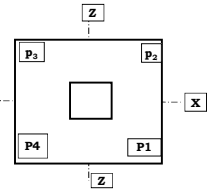
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² (BH28) For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	Base Pressure (T/m²) V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	Remarks
DL+LL	18.18	0.08	0.00	5.31	0.08	0.00	7.39	7.39	7.23	7.23	5.23	O.K
DL+0.75*LL+0.75*WL1(+)x	20.02	3.40	0.00	5.85	3.22	0.00	11.07	11.07	4.63	4.63	8.91	O.K
DL+0.75*LL+0.75*WL2(+)x	17.70	3.31	0.00	5.17	3.14	0.00	10.31	10.31	4.03	4.03	8.15	O.K
DL+0.75*LL+0.75*WL3(-)x	16.52	3.23	0.00	4.83	3.07	0.00	9.89	9.89	3.76	3.76	7.73	O.K
DL+0.75*LL+0.75*WL4(-)x	14.25	3.21	0.00	4.16	3.04	0.00	9.20	9.20	3.13	3.13	7.04	O.K
DL+0.75*LL+0.75*WL5(+)z	13.01	0.02	1.19	3.80	0.02	1.13	6.95	4.69	6.91	4.65	4.79	O.K
DL+0.75*LL+0.75*WL6(+)z	12.74	0.03	1.19	3.72	0.02	1.13	6.87	4.62	6.82	4.57	4.71	O.K
DL+0.75*LL+0.75*WL7(-)z	13.01	0.02	1.19	3.80	0.02	1.13	6.95	4.69	6.91	4.65	4.79	O.K
DL+0.75*LL+0.75*WL8(-)z	12.74	0.03	1.19	3.72	0.02	1.13	6.87	4.62	6.82	4.57	4.71	O.K
0.6*DL+WL1(+)x	16.63	4.44	0.00	4.86	4.21	0.00	11.07	11.07	2.65	2.65	8.91	O.K
0.6*DL+WL2(+)x	13.54	4.33	0.00	3.96	4.10	0.00	10.06	10.06	1.85	1.85	7.90	O.K
0.6*DL+WL3(-)x	11.97	4.40	0.00	3.50	4.17	0.00	9.67	9.67	1.33	1.33	7.51	O.K
0.6*DL+WL4(-)x	8.94	4.36	0.00	2.61	4.13	0.00	8.74	8.74	0.48	0.48	6.58	O.K
0.6*DL+WL5(+)z	7.28	0.11	1.59	2.13	0.11	1.50	5.74	2.73	5.52	2.52	3.58	O.K
0.6*DL+WL6(+)z	6.92	0.12	1.59	2.02	0.11	1.50	5.64	2.63	5.41	2.40	3.48	O.K
0.6*DL+WL7(-)z	7.28	0.11	1.59	2.13	0.11	1.50	5.74	2.73	5.52	2.52	3.58	O.K
0.6*DL+WL8(-)z	6.92	0.12	1.59	2.02	0.11	1.50	5.64	2.63	5.41	2.40	3.48	O.K

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

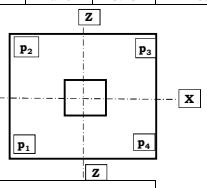
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)
1.4 DL	16.1	0.01	0.00	4.72	0.01	0.00	6.72	6.72	6.71	6.71	4.02
1.4 DL + 1.7 LL	22.3	0.14	0.00	6.52	0.14	0.00	8.65	6.65	6.38	6.38	5.95
1.2 DL + 1.6 WL1(+)x + LL	24.8	7.20	0.00	7.25	6.82	0.00	16.07	14.07	0.44	0.44	13.37
1.2 DL + 1.6 WL2(+)x + LL	17.5	7.01	0.00	5.11	6.65	0.00	13.76	11.76	-1.54	-1.54	11.06
1.2 DL + 1.6 WL3(-)x + LL	17.4	6.95	0.00	5.08	6.59	0.00	13.66	11.66	-1.51	-1.51	10.96
1.2 DL + 1.6 WL4(-)x + LL	12.5	6.89	0.00	3.66	6.53	0.00	12.19	10.19	-2.87	-2.87	9.49
1.2 DL + 1.6 WL5(+)z + LL	9.9	0.09	2.54	2.89	0.09	2.41	7.38	0.57	5.20	0.39	4.68
1.2 DL + 1.6 WL6(+)z + LL	9.3	0.10	2.54	2.72	0.10	2.41	7.22	0.41	5.02	0.21	4.52
1.2 DL + 1.6 WL7(-)z + LL	9.9	0.09	2.54	2.89	0.09	2.41	7.38	0.57	5.20	0.39	4.68
1.2 DL + 1.6 WL8(-)z + LL	9.3	0.10	2.54	2.72	0.10	2.41	7.22	0.41	5.02	0.21	4.52
0.9 DL + 1.6 WL1(+)x	20.0	7.11	0.00	5.85	6.74	0.00	14.58	12.58	-0.89	-0.89	11.88
0.9 DL + 1.6 WL2(+)x	15.1	6.93	0.00	4.40	6.56	0.00	12.97	10.97	-2.16	-2.16	10.27
0.9 DL + 1.6 WL3(-)x	12.6	7.04	0.00	3.67	6.67	0.00	12.34	10.34	-3.00	-3.00	9.64
0.9 DL + 1.6 WL4(-)x	7.7	6.98	0.00	2.25	6.61	0.00	10.86	8.86	-4.36	-4.36	8.16
0.9 DL + 1.6 WL5(+)z	5.1	0.18	2.54	1.48	0.17	2.41	6.06	-0.75	3.71	-1.10	3.36
0.9 DL + 1.6 WL6(+)z	4.5	0.19	2.54	1.31	0.18	2.41	5.90	-0.91	3.53	-1.28	3.20
0.9 DL + 1.6 WL7(-)z	5.1	0.18	2.54	1.48	0.17	2.41	6.06	-0.75	3.71	-1.10	3.36
0.9 DL + 1.6 WL8(-)z	4.5	0.19	2.54	1.31	0.18	2.41	5.90	-0.91	3.53	-1.28	3.20

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-05-2020	29-06-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
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Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2

= 3.004 m

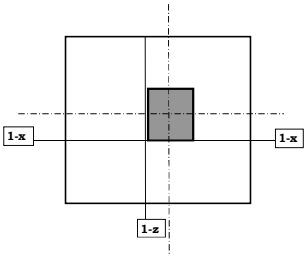
Punching Area= ((0.55+2*0.5*0.201)*(0.55+2*0.5*0.201))

= 0.56 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)

= 2.91 T/m^2

Area of footing, A_f= 3.42 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 38.27 T

Strength of Concrete in shear Vc1 = 129.14 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 139.21 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 119.07 T **Table 22.6.5.2_ (c)**

Design Vc= 119.07 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (38.27*10000)/(3004*201)

= 0.63 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$	$S.F_{(Oneway)}$ (T)	$S.F_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear ($0.75*\sqrt{fc}/6$ (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4 DL	4.02	0.85	1.74	0.85	1.80	11.51	0.19	OK	0.74	0.09	OK	0.21
1.4 DL + 1.7 LL	5.95	1.26	2.58	1.26	2.67	17.03	0.28	OK	0.74	0.133	OK	0.31
1.2 DL + 1.6 WL1(+)x + LL	13.37	2.82	5.79	2.82	6.00	38.27	0.63	OK	0.74	0.299	OK	0.70
1.2 DL + 1.6 WL2(+)x + LL	11.06	2.34	4.79	2.34	4.97	31.66	0.52	OK	0.74	0.247	OK	0.58
1.2 DL + 1.6 WL3(-)x + LL	10.96	2.32	4.75	2.32	4.92	31.37	0.52	OK	0.74	0.245	OK	0.57
1.2 DL + 1.6 WL4(-)x + LL	9.49	2.00	4.11	2.00	4.26	27.17	0.45	OK	0.74	0.212	OK	0.50
1.2 DL + 1.6 WL5(+)z + LL	4.68	0.99	2.03	0.99	2.10	13.40	0.22	OK	0.74	0.105	OK	0.24
1.2 DL + 1.6 WL6(+)z + LL	4.52	0.95	1.96	0.95	2.03	12.94	0.21	OK	0.74	0.101	OK	0.24
1.2 DL + 1.6 WL7(-)z + LL	4.68	0.99	2.03	0.99	2.10	13.40	0.22	OK	0.74	0.105	OK	0.24
1.2 DL + 1.6 WL8(-)z + LL	4.52	0.95	1.96	0.95	2.03	12.94	0.21	OK	0.74	0.101	OK	0.24
0.9 DL + 1.6 WL1(+)x	11.88	2.51	5.14	2.51	5.33	34.01	0.56	OK	0.74	0.265	OK	0.62
0.9 DL + 1.6 WL2(+)x	10.27	2.17	4.45	2.17	4.61	29.40	0.49	OK	0.74	0.229	OK	0.54
0.9 DL + 1.6 WL3(-)x	9.64	2.04	4.17	2.04	4.33	27.59	0.46	OK	0.74	0.215	OK	0.50
0.9 DL + 1.6 WL4(-)x	8.16	1.72	3.53	1.72	3.66	23.36	0.39	OK	0.74	0.182	OK	0.43
0.9 DL + 1.6 WL5(+)z	3.36	0.71	1.45	0.71	1.51	9.62	0.16	OK	0.74	0.075	OK	0.18
0.9 DL + 1.6 WL6(+)z	3.20	0.68	1.39	0.68	1.44	9.16	0.15	OK	0.74	0.071	OK	0.17
0.9 DL + 1.6 WL7(-)z	3.36	0.71	1.45	0.71	1.51	9.62	0.16	OK	0.74	0.075	OK	0.18
0.9 DL + 1.6 WL8(-)z	3.20	0.68	1.39	0.68	1.44	9.16	0.15	OK	0.74	0.071	OK	0.17

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4 DL	0.28	0.001	217
1.4 DL + 1.7 LL	0.41	0.001	217
1.2 DL + 1.6 WL1(+)x + LL	0.93	0.002	434
1.2 DL + 1.6 WL2(+)x + LL	0.77	0.002	434
1.2 DL + 1.6 WL3(-)x + LL	0.76	0.002	434
1.2 DL + 1.6 WL4(-)x + LL	0.66	0.002	434
1.2 DL + 1.6 WL5(+)z + LL	0.33	0.001	217
1.2 DL + 1.6 WL6(+)z + LL	0.32	0.001	217
1.2 DL + 1.6 WL7(-)z + LL	0.33	0.001	217
1.2 DL + 1.6 WL8(-)z + LL	0.32	0.001	217
0.9 DL + 1.6 WL1(+)x	0.83	0.002	434
0.9 DL + 1.6 WL2(+)x	0.72	0.002	434
0.9 DL + 1.6 WL3(-)x	0.67	0.002	434
0.9 DL + 1.6 WL4(-)x	0.57	0.001	217
0.9 DL + 1.6 WL5(+)z	0.23	0.001	217
0.9 DL + 1.6 WL6(+)z	0.22	0.001	217
0.9 DL + 1.6 WL7(-)z	0.23	0.001	217
0.9 DL + 1.6 WL8(-)z	0.22	0.001	217

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing	Spacing provd.
At bottom	2.82	0.70	0.93	0.002	434.00	390.60	12ϕT	260.59	300	200
At top	0.00	0.00	0.00	0.0018	361.80	361.80	12ϕT	312.60	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**12ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-05-2020	29-06-2020	26-10-2023
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3.2 Design of Foundation mkd. F2 (GRID A & D- 2,3, 4, 5 6)

	A (2,3,4,5,6)		
LOAD DEFINATION	Fx Ton	Fy Ton	Fz Ton
1 DL	0.0	1.8	0.0
2 LLROOF	0.0	1.8	0.0
21 WL1(+X)CASE-A	0.5	2.1	0.0
22 WL2(+X)CASE-B	0.5	0.5	0.0
23 WL3(-X)CASE-A	-0.5	-0.7	0.0
24 WL4(-X)CASE-B	-0.5	-2.2	0.0
25 WL5(+Z)CASE-A	0.0	-2.9	1.0
26 WL6(+Z)CASE-B	0.0	-3.1	1.0
27 WL7(-Z)CASE-A	0.0	-2.9	-1.0
28 WL8(-Z)CASE-B	0.0	-3.1	-1.0

STRENGTH DESIGN	GRID A (2,3,4,5,6)		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4 DL	0.00	2.51	0.00
1.4 DL + 1.7 LL	0.00	5.50	0.00
1.2 DL + 1.6 WL1(+)x + LL	0.82	7.20	0.00
1.2 DL + 1.6 WL2(+)x + LL	0.82	3.25	0.00
1.2 DL + 1.6 WL3(-)x + LL	-0.82	2.79	0.00
1.2 DL + 1.6 WL4(-)x + LL	-0.82	0.33	0.00
1.2 DL + 1.6 WL5(+)z + LL	0.00	-0.72	1.64
1.2 DL + 1.6 WL6(+)z + LL	0.00	-1.02	1.64
1.2 DL + 1.6 WL7(-)z + LL	0.00	-0.72	-1.64
1.2 DL + 1.6 WL8(-)z + LL	0.00	-1.02	-1.64
0.9 DL + 1.6 WL1(+)x	0.82	4.90	0.00
0.9 DL + 1.6 WL2(+)x	0.82	2.34	0.00
0.9 DL + 1.6 WL3(-)x	-0.82	0.49	0.00
0.9 DL + 1.6 WL4(-)x	-0.82	-1.97	0.00
0.9 DL + 1.6 WL5(+)z	0.00	-3.02	1.64
0.9 DL + 1.6 WL6(+)z	0.00	-3.32	1.64
0.9 DL + 1.6 WL7(-)z	0.00	-3.02	-1.64
0.9 DL + 1.6 WL8(-)z	0.00	-3.32	-1.64

Global Project ID	900707
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Ma'aden Revision No:	A01	C01	D01
Date	29-05-2020	29-06-2020	26-10-2023
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DESIGN OF FOUNDATION MKD.F2

	About-X	About-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.65m	1.65m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

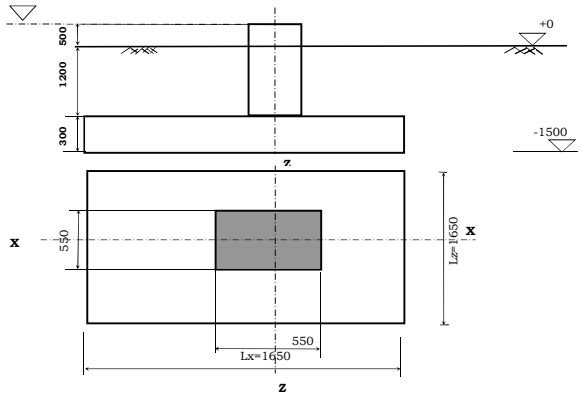
Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.65*1.65*0.3*2.5)+(0.55*0.55*1.7*2.5) = 3.33T

Weight of soil = ((1.65*1.65)-(0.55*0.55))*1.2*1.8 = 5.23T

U/S Base Pl Total = 8.55T



Design Input:

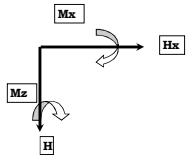
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = ######## 0.215m

Effective depth of slab along Z direction = ######## 0.195m

Clear projection along Z direction = 0.55m

Clear projection along X direction = 0.55m



WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.00	3.55	0.00	0.0	0.00	12.10	0.00	0.00	0.00	0.00
DL+0.75*LL+0.75*WL1(+)x	0.38	4.65	0.00	0.0	0.00	13.20	0.38	0.00	0.00	0.77
DL+0.75*LL+0.75*WL2(+)x	0.38	3.45	0.00	0.0	0.00	12.00	0.38	0.00	0.00	0.77
DL+0.75*LL+0.75*WL3(-)x	-0.38	2.59	0.00	0.0	0.00	11.14	-0.38	0.00	0.00	0.77
DL+0.75*LL+0.75*WL4(-)x	-0.38	1.43	0.00	0.0	0.00	9.98	-0.38	0.00	0.00	0.77
DL+0.75*LL+0.75*WL5(+)z	0.00	0.94	0.77	0.0	0.00	9.49	0.00	1.54	0.77	0.00
DL+0.75*LL+0.75*WL6(+)z	0.00	0.80	0.77	0.0	0.00	9.35	0.00	1.54	0.77	0.00
DL+0.75*LL+0.75*WL7(-)z	0.00	0.94	-0.77	0.0	0.00	9.49	0.00	1.54	-0.77	0.00
DL+0.75*LL+0.75*WL8(-)z	0.00	0.80	-0.77	0.0	0.00	9.35	0.00	1.54	-0.77	0.00
0.6*DL+WL1(+)x	0.51	3.13	0.00	0.0	0.00	11.68	0.51	0.00	0.00	1.02
0.6*DL+WL2(+)x	0.51	1.53	0.00	0.0	0.00	10.08	0.51	0.00	0.00	1.02
0.6*DL+WL3(-)x	-0.51	0.37	0.00	0.0	0.00	8.92	-0.51	0.00	0.00	1.02
0.6*DL+WL4(-)x	-0.51	-1.16	0.00	0.0	0.00	7.39	-0.51	0.00	0.00	1.02
0.6*DL+WL5(+)z	0.00	-1.82	1.03	0.0	0.00	6.73	0.00	2.05	1.03	0.00
0.6*DL+WL6(+)z	0.00	-2.00	1.03	0.0	0.00	6.55	0.00	2.05	1.03	0.00
0.6*DL+WL7(-)z	0.00	-1.82	-1.03	0.0	0.00	6.73	0.00	2.05	-1.03	0.00
0.6*DL+WL8(-)z	0.00	-2.00	-1.03	0.0	0.00	6.55	0.00	2.05	-1.03	0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-05-2020	29-06-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	29-05-2020	29-06-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+LL	12.10	0.00	0.00	0.00	0.00	NA	NA	NA	NA
DL+0.75*LL+0.75*WL1(+)x	13.20	0.38	0.77	0.00	0.00	15.97	NA	13.81	NA
DL+0.75*LL+0.75*WL2(+)x	12.00	0.38	0.77	0.00	0.00	14.51	NA	12.55	NA
DL+0.75*LL+0.75*WL3(-)x	11.14	-0.38	0.77	0.00	0.00	13.47	NA	11.65	NA
DL+0.75*LL+0.75*WL4(-)x	9.98	-0.38	0.77	0.00	0.00	12.07	NA	10.44	NA
DL+0.75*LL+0.75*WL5(+)z	9.49	0.00	0.00	0.77	1.54	NA	5.70	NA	4.93
DL+0.75*LL+0.75*WL6(+)z	9.35	0.00	0.00	0.77	1.54	NA	5.62	NA	4.86
DL+0.75*LL+0.75*WL7(-)z	9.49	0.00	0.00	-0.77	1.54	NA	5.70	NA	4.93
DL+0.75*LL+0.75*WL8(-)z	9.35	0.00	0.00	-0.77	1.54	NA	5.62	NA	4.86
0.6*DL+WL1(+)x	11.68	0.51	1.02	0.00	0.00	10.59	NA	9.16	NA
0.6*DL+WL2(+)x	10.08	0.51	1.02	0.00	0.00	9.14	NA	7.90	NA
0.6*DL+WL3(-)x	8.92	-0.51	1.02	0.00	0.00	8.09	NA	7.00	NA
0.6*DL+WL4(-)x	7.39	-0.51	1.02	0.00	0.00	6.70	NA	5.79	NA
0.6*DL+WL5(+)z	6.73	0.00	0.00	1.03	2.05	NA	3.03	NA	2.62
0.6*DL+WL6(+)z	6.55	0.00	0.00	1.03	2.05	NA	2.95	NA	2.55
0.6*DL+WL7(-)z	6.73	0.00	0.00	-1.03	2.05	NA	3.03	NA	2.62
0.6*DL+WL8(-)z	6.55	0.00	0.00	-1.03	2.05	NA	2.95	NA	2.55

ULTIMATE STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4 DL	0.00	2.51	0.00	0.0	0.00	11.057	0.00	0.00
1.4 DL + 1.7 LL	0.00	5.50	0.00	0.0	0.00	14.054	0.00	0.00
1.2 DL + 1.6 WL1(+)x + LL	0.82	7.20	0.00	0.0	0.00	15.750	0.00	1.63
1.2 DL + 1.6 WL2(+)x + LL	0.82	3.25	0.00	0.0	0.00	11.798	0.00	1.63
1.2 DL + 1.6 WL3(-)x + LL	-0.82	2.79	0.00	0.0	0.00	11.341	0.00	1.63
1.2 DL + 1.6 WL4(-)x + LL	-0.82	0.33	0.00	0.0	0.00	8.879	0.00	1.63
1.2 DL + 1.6 WL5(+)z + LL	0.00	-0.72	1.64	0.0	0.00	7.830	3.28	0.00
1.2 DL + 1.6 WL6(+)z + LL	0.00	-1.02	1.64	0.0	0.00	7.535	3.28	0.00
1.2 DL + 1.6 WL7(-)z + LL	0.00	-0.72	-1.64	0.0	0.00	7.830	3.28	0.00
1.2 DL + 1.6 WL8(-)z + LL	0.00	-1.02	-1.64	0.0	0.00	7.535	3.28	0.00
0.9 DL + 1.6 WL1(+)x	0.82	4.90	0.00	0.0	0.00	13.450	0.00	1.63
0.9 DL + 1.6 WL2(+)x	0.82	2.34	0.00	0.0	0.00	10.887	0.00	1.63
0.9 DL + 1.6 WL3(-)x	-0.82	0.49	0.00	0.0	0.00	9.041	0.00	1.63
0.9 DL + 1.6 WL4(-)x	-0.82	-1.97	0.00	0.0	0.00	6.579	0.00	1.63
0.9 DL + 1.6 WL5(+)z	0.00	-3.02	1.64	0.0	0.00	5.530	3.28	0.00
0.9 DL + 1.6 WL6(+)z	0.00	-3.32	1.64	0.0	0.00	5.235	3.28	0.00
0.9 DL + 1.6 WL7(-)z	0.00	-3.02	-1.64	0.0	0.00	5.530	3.28	0.00
0.9 DL + 1.6 WL8(-)z	0.00	-3.32	-1.64	0.0	0.00	5.235	3.28	0.00

CHECK FOR BEARING PRESSURE

Load Combination & Calculation of Base pressure

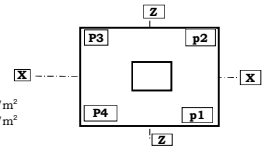
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Load Case	Design Load/moments			Base Pressure (T/m²)								
	V (T)	Mz (T-m)	Mx (T-m)	V/(Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/ Lx.(Lz)²	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
DL+LL	12.10	0.00	0.00	4.45	0.00	0.00	6.45	6.45	6.45	6.45	4.29	O.K
DL+0.75*LL+0.75*WL1(+)x	13.20	0.77	0.00	4.85	1.02	0.00	7.87	7.87	5.83	5.83	5.71	O.K
DL+0.75*LL+0.75*WL2(+)x	12.00	0.77	0.00	4.41	1.02	0.00	7.43	7.43	5.39	5.39	5.27	O.K
DL+0.75*LL+0.75*WL3(-)x	11.14	0.77	0.00	4.09	1.02	0.00	7.11	7.11	5.07	5.07	4.95	O.K
DL+0.75*LL+0.75*WL4(-)x	9.98	0.77	0.00	3.67	1.02	0.00	6.69	6.69	4.65	4.65	4.53	O.K
DL+0.75*LL+0.75*WL5(+)z	9.49	0.00	1.54	3.49	0.00	2.06	7.54	3.43	7.54	3.43	5.38	O.K
DL+0.75*LL+0.75*WL6(+)z	9.35	0.00	1.54	3.44	0.00	2.06	7.49	3.38	7.49	3.38	5.33	O.K
DL+0.75*LL+0.75*WL7(-)z	9.49	0.00	1.54	3.49	0.00	2.06	7.54	3.43	7.54	3.43	5.38	O.K
DL+0.75*LL+0.75*WL8(-)z	9.35	0.00	1.54	3.44	0.00	2.06	7.49	3.38	7.49	3.38	5.33	O.K
0.6*DL+WL1(+)x	11.68	1.02	0.00	4.29	1.36	0.00	7.65	7.65	4.93	4.93	5.49	O.K
0.6*DL+WL2(+)x	10.08	1.02	0.00	3.70	1.36	0.00	7.06	7.06	4.34	4.34	4.90	O.K
0.6*DL+WL3(-)x	8.92	1.02	0.00	3.28	1.36	0.00	6.64	6.64	3.92	3.92	4.48	O.K
0.6*DL+WL4(-)x	7.39	1.02	0.00	2.71	1.36	0.00	6.08	6.08	3.35	3.35	3.92	O.K
0.6*DL+WL5(+)z	6.73	0.00	2.05	2.47	0.00	2.74	7.21	1.73	7.21	1.73	5.05	O.K
0.6*DL+WL6(+)z	6.55	0.00	2.05	2.40	0.00	2.74	7.14	1.66	7.14	1.66	4.98	O.K
0.6*DL+WL7(-)z	6.73	0.00	2.05	2.47	0.00	2.74	7.21	1.73	7.21	1.73	5.05	O.K
0.6*DL+WL8(-)z	6.55	0.00	2.05	2.40	0.00	2.74	7.14	1.66	7.14	1.66	4.98	O.K

CALCULATION OF DESIGN PRESSURE

Load Combination & Calculation of Base pressure

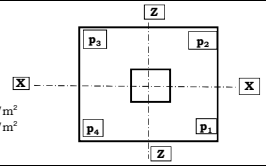
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-05-2020	29-06-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	29-05-2020	29-06-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	Base Pressure (T/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)
1.4 DL	11.1	0.00	0.00	4.06	0.00	0.00	6.06	6.06	6.06	6.06	3.36
1.4 DL + 1.7 LL	14.1	0.00	0.00	5.16	0.00	0.00	7.16	7.16	7.16	7.16	4.46
1.2 DL + 1.6 WL1(+)x + LL	15.7	1.63	0.00	5.79	2.18	0.00	9.96	9.96	5.61	5.61	7.26
1.2 DL + 1.6 WL2(+)x + LL	11.8	1.63	0.00	4.33	2.18	0.00	8.51	8.51	4.15	4.15	5.81
1.2 DL + 1.6 WL3(-)x + LL	11.3	1.63	0.00	4.17	2.18	0.00	8.35	8.35	3.99	3.99	5.65
1.2 DL + 1.6 WL4(-)x + LL	8.9	1.63	0.00	3.26	2.18	0.00	7.44	7.44	3.08	3.08	4.74
1.2 DL + 1.6 WL5(+)z + LL	7.8	0.00	3.28	2.88	0.00	4.39	9.26	0.49	9.26	0.49	6.56
1.2 DL + 1.6 WL6(+)z + LL	7.5	0.00	3.28	2.77	0.00	4.39	9.15	0.38	9.15	0.38	6.45
1.2 DL + 1.6 WL7(-)z + LL	7.8	0.00	3.28	2.88	0.00	4.39	9.26	0.49	9.26	0.49	6.56
1.2 DL + 1.6 WL8(-)z + LL	7.5	0.00	3.28	2.77	0.00	4.39	9.15	0.38	9.15	0.38	6.45
0.9 DL + 1.6 WL1(+)x	13.4	1.63	0.00	4.94	2.18	0.00	9.12	9.12	4.76	4.76	6.42
0.9 DL + 1.6 WL2(+)x	10.9	1.63	0.00	4.00	2.18	0.00	8.18	8.18	3.82	3.82	5.48
0.9 DL + 1.6 WL3(-)x	9.0	1.63	0.00	3.32	2.18	0.00	7.50	7.50	3.14	3.14	4.80
0.9 DL + 1.6 WL4(-)x	6.6	1.63	0.00	2.42	2.18	0.00	6.60	6.60	2.24	2.24	3.90
0.9 DL + 1.6 WL5(+)z	5.5	0.00	3.28	2.03	0.00	4.39	8.42	-0.35	8.42	-0.35	5.72
0.9 DL + 1.6 WL6(+)z	5.2	0.00	3.28	1.92	0.00	4.39	8.31	-0.46	8.31	-0.46	5.61
0.9 DL + 1.6 WL7(-)z	5.5	0.00	3.28	2.03	0.00	4.39	8.42	-0.35	8.42	-0.35	5.72
0.9 DL + 1.6 WL8(-)z	5.2	0.00	3.28	1.92	0.00	4.39	8.31	-0.46	8.31	-0.46	5.61

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM1z =Moment at the face of column 1z-1Z in T-m/m.

SF(oneway) = Oneway shear force at a distance 'd' from the face of column in T/m.

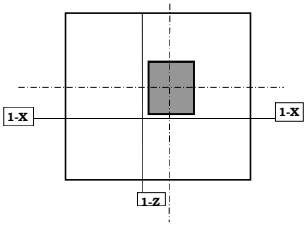
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2

= 2.98 m

Punching Area= ((0.55+2*0.5*0.195)*(0.55+2*0.5*0.195))

= 0.56 m^2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)

= 2.91 T/m^2

Area of footing, A_f= 2.72 m^2

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 15.70 T

Strength of Concrete in shear Vc1 = 124.28 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 132.28 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 114.59 T **Table 22.6.5.2_ (c)**

Design Vc= 114.59 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (15.7*10000)/(2980*195)

= 0.27 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$ ™	$S.F_{(oneway)}$ (T)	$S.F_{(punching)}$ (T)	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4 DL	3.36	0.51	1.13	0.51	1.19	7.27	0.13	OK	0.74	0.061	OK	0.13
1.4 DL + 1.7 LL	4.46	0.67	1.49	0.67	1.58	9.64	0.17	OK	0.74	0.081	OK	0.18
1.2 DL + 1.6 WL1(+)x + LL	7.26	1.10	2.43	1.10	2.58	15.70	0.27	OK	0.74	0.132	OK	0.29
1.2 DL + 1.6 WL2(+)x + LL	5.81	0.88	1.95	0.88	2.06	12.56	0.22	OK	0.74	0.106	OK	0.23
1.2 DL + 1.6 WL3(-)x + LL	5.65	0.85	1.89	0.85	2.01	12.22	0.21	OK	0.74	0.103	OK	0.22
1.2 DL + 1.6 WL4(-)x + LL	4.74	0.72	1.59	0.72	1.68	10.25	0.18	OK	0.74	0.086	OK	0.19
1.2 DL + 1.6 WL5(+)z + LL	6.56	0.99	2.20	0.99	2.33	14.19	0.24	OK	0.74	0.119	OK	0.26
1.2 DL + 1.6 WL6(+)z + LL	6.45	0.98	2.16	0.98	2.29	13.95	0.24	OK	0.74	0.117	OK	0.26
1.2 DL + 1.6 WL7(-)z + LL	6.56	0.99	2.20	0.99	2.33	14.19	0.24	OK	0.74	0.119	OK	0.26
1.2 DL + 1.6 WL8(-)z + LL	6.45	0.98	2.16	0.98	2.29	13.95	0.24	OK	0.74	0.117	OK	0.26
0.9 DL + 1.6 WL1(+)x	6.42	0.97	2.15	0.97	2.28	13.88	0.24	OK	0.74	0.117	OK	0.26
0.9 DL + 1.6 WL2(+)x	5.48	0.83	1.84	0.83	1.95	11.85	0.2	OK	0.74	0.1	OK	0.22
0.9 DL + 1.6 WL3(-)x	4.80	0.73	1.61	0.73	1.70	10.38	0.18	OK	0.74	0.087	OK	0.19
0.9 DL + 1.6 WL4(-)x	3.90	0.59	1.31	0.59	1.38	8.43	0.15	OK	0.74	0.071	OK	0.16
0.9 DL + 1.6 WL5(+)z	5.72	0.87	1.92	0.87	2.03	12.37	0.21	OK	0.74	0.104	OK	0.23
0.9 DL + 1.6 WL6(+)z	5.61	0.85	1.88	0.85	1.99	12.13	0.21	OK	0.74	0.102	OK	0.22
0.9 DL + 1.6 WL7(-)z	5.72	0.87	1.92	0.87	2.03	12.37	0.21	OK	0.74	0.104	OK	0.23
0.9 DL + 1.6 WL8(-)z	5.61	0.85	1.88	0.85	1.99	12.13	0.21	OK	0.74	0.102	OK	0.22

Load Case	kn (ϕ=0.75)	ρ =	A_{st}
Ultimate Conditions			
1.4 DL	0.18	0	0
1.4 DL + 1.7 LL	0.24	0.001	215
1.2 DL + 1.6 WL1(+)x + LL	0.39	0.001	215
1.2 DL + 1.6 WL2(+)x + LL	0.31	0.001	215
1.2 DL + 1.6 WL3(-)x + LL	0.3	0.001	215
1.2 DL + 1.6 WL4(-)x + LL	0.25	0.001	215
1.2 DL + 1.6 WL5(+)z + LL	0.35	0.001	215
1.2 DL + 1.6 WL6(+)z + LL	0.34	0.001	215
1.2 DL + 1.6 WL7(-)z + LL	0.35	0.001	215
1.2 DL + 1.6 WL8(-)z + LL	0.34	0.001	215
0.9 DL + 1.6 WL1(+)x	0.34	0.001	215
0.9 DL + 1.6 WL2(+)x	0.29	0.001	215
0.9 DL + 1.6 WL3(-)x	0.25	0.001	215
0.9 DL + 1.6 WL4(-)x	0.21	0.001	215
0.9 DL + 1.6 WL5(+)z	0.3	0.001	215
0.9 DL + 1.6 WL6(+)z	0.3	0.001	215
0.9 DL + 1.6 WL7(-)z	0.3	0.001	215
0.9 DL + 1.6 WL8(-)z	0.3	0.001	215

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)
At bottom	1.10	0.29	0.39	0.001	215.00	387.00	12ϕT	292.24	300	200
At top	0.00	0.00	0.00	0.0018	351.00	351.00	12ϕT	322.21	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **200** c/c At Top **12ϕT** **200** c/c

Along Z Direction :

At Bottom **12ϕT** **200** c/c At Top **12ϕT** **200** c/c

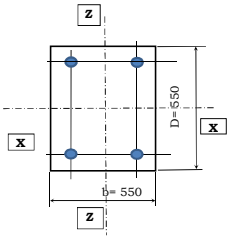
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

DESIGN OF COLUMN MKD C1

Input:	
Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	1700

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsuported length of column about X (l_{ux}) = 1700.00
Unsupported length of column about Z (l_{uz}) = 1700.00

For Columns not braced against sidesway,
Slenderness ratio, $K\,l_{ux} / r_x$ = 21.42 < 22 SECTION 6.2.5a
$K\,l_{uz} / r_z$ = 21.42 < 22 SECTION 6.2.5a

Where, Effective length factor K= 2
Radius of gyration r_x= 159
Radius of gyration r_z= 159

γ_x = 0.68
γ_z = 0.68

Load Combination (LOAD OF COL. B7)	Factored Load/Moment (T)			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
1.4 DL	0.00	5.69	0.02	0.0	8.8	0.1	0.01
1.4 DL + 1.7 LL	0.01	9.23	0.02	0.0	14.2	0.1	0.01
1.2 DL + 1.6 WL1(+)x + LL	0.01	3.99	0.02	0.0	6.1	0.1	0.01
1.2 DL + 1.6 WL2(+)x + LL	0.00	3.32	0.02	0.0	5.1	0.0	0.00
1.2 DL + 1.6 WL3(-)x + LL	0.01	12.34	0.02	0.0	19.0	0.1	0.02
1.2 DL + 1.6 WL4(-)x + LL	0.01	9.14	0.02	0.0	14.1	0.0	0.01
1.2 DL + 1.6 WL5(+)z + LL	0.32	2.70	5.91	0.8	4.2	15.5	0.00
1.2 DL + 1.6 WL6(+)z + LL	0.34	2.38	5.91	0.9	3.7	15.5	0.00
1.2 DL + 1.6 WL7(-)z + LL	0.31	0.91	-5.88	0.8	1.4	15.4	0.00
1.2 DL + 1.6 WL8(-)z + LL	0.33	0.58	-5.88	0.9	0.9	15.4	0.00
0.9 DL + 1.6 WL1(+)x	0.00	0.69	0.02	0.0	1.1	0.0	0.00
0.9 DL + 1.6 WL2(+)x	0.00	-2.08	0.01	0.0	-3.2	0.0	0.00
0.9 DL + 1.6 WL3(-)x	0.01	9.04	0.01	0.0	13.9	0.0	0.01
0.9 DL + 1.6 WL4(-)x	0.00	5.84	0.01	0.0	9.0	0.0	0.01
0.9 DL + 1.6 WL5(+)z	0.32	-0.60	5.90	0.8	-0.9	15.4	0.00
0.9 DL + 1.6 WL6(+)z	0.34	-0.93	5.90	0.9	-1.4	15.4	0.00
0.9 DL + 1.6 WL7(-)z	0.31	-2.39	-5.88	0.8	-3.7	15.4	0.00
0.9 DL + 1.6 WL8(-)z	0.33	-2.72	-5.88	0.9	-4.2	15.4	0.00

REINFORCEMENT

Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)	A_{sc} pvd. =	3221.43 Sq mm
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)		
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)	Total A_{sc} =	1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 1025.65375 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.7		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cbD^2$						
1.4 DL	0.06	0.06	34.9	34.94	0.55	1.16	0.00	O.K
1.4 DL + 1.7 LL	0.07	0.07	40.8	40.76	0.55	1.16	0.00	O.K
1.2 DL + 1.6 WL1(+)x + LL	0.06	0.06	34.9	34.94	0.55	1.16	0.00	O.K
1.2 DL + 1.6 WL2(+)x + LL	0.06	0.06	34.9	34.94	0.55	1.16	0.00	O.K
1.2 DL + 1.6 WL3(-)x + LL	0.06	0.06	34.9	34.94	0.55	1.16	0.00	O.K
1.2 DL + 1.6 WL4(-)x + LL	0.06	0.06	34.9	34.94	0.55	1.16	0.00	O.K
1.2 DL + 1.6 WL5(+)z + LL	0.06	0.06	34.9	34.94	0.55	1.16	0.40	O.K
1.2 DL + 1.6 WL6(+)z + LL	0.06	0.06	34.9	34.94	0.55	1.16	0.40	O.K
1.2 DL + 1.6 WL7(-)z + LL	0.06	0.06	34.9	34.94	0.55	1.16	0.40	O.K
1.2 DL + 1.6 WL8(-)z + LL	0.06	0.06	34.9	34.94	0.55	1.16	0.40	O.K
0.9 DL + 1.6 WL1(+)x	0.06	0.06	34.9	34.94	0.55	1.16	0.00	O.K
0.9 DL + 1.6 WL2(+)x	0.06	0.06	34.9	34.94	0.55	1.16	0.00	O.K
0.9 DL + 1.6 WL3(-)x	0.06	0.06	34.9	34.94	0.55	1.16	0.00	O.K
0.9 DL + 1.6 WL4(-)x	0.06	0.06	34.9	34.94	0.55	1.16	0.00	O.K
0.9 DL + 1.6 WL5(+)z	0.06	0.06	34.9	34.94	0.55	1.16	0.40	O.K
0.9 DL + 1.6 WL6(+)z	0.06	0.06	34.9	34.94	0.55	1.16	0.40	O.K
0.9 DL + 1.6 WL7(-)z	0.06	0.06	34.9	34.94	0.55	1.16	0.40	O.K
0.9 DL + 1.6 WL8(-)z	0.06	0.06	34.9	34.94	0.55	1.16	0.40	O.K

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3.3 Design of Foundation mkd. F3 (GRID A,B,C,D - 1 & 7)

	B 7		
LOAD DEFINATION	Fx Ton	Fy Ton	Fz Ton
1 DL	0.0	4.1	0.0
2 LLROOF	0.0	2.1	0.0
21 WL1(+X)CASE-A	0.0	-1.9	0.0
22 WL2(+X)CASE-B	0.0	-3.6	0.0
23 WL3(-X)CASE-A	0.0	3.4	0.0
24 WL4(-X)CASE-B	0.0	1.4	0.0
25 WL5(+Z)CASE-A	0.2	-2.7	3.7
26 WL6(+Z)CASE-B	0.2	-2.9	3.7
27 WL7(-Z)CASE-A	0.2	-3.8	-3.7
28 WL8(-Z)CASE-B	0.2	-4.0	-3.7

STRENGTH DESIGN	B- 7		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4 DL	0.00	5.69	0.02
1.4 DL + 1.7 LL	0.01	9.23	0.02
1.2 DL + 1.6 WL1(+)x + LL	0.01	3.99	0.02
1.2 DL + 1.6 WL2(+)x + LL	0.00	3.32	0.02
1.2 DL + 1.6 WL3(-)x + LL	0.01	12.34	0.02
1.2 DL + 1.6 WL4(-)x + LL	0.01	9.14	0.02
1.2 DL + 1.6 WL5(+)z + LL	0.32	2.70	5.91
1.2 DL + 1.6 WL6(+)z + LL	0.34	2.38	5.91
1.2 DL + 1.6 WL7(-)z + LL	0.31	0.91	-5.88
1.2 DL + 1.6 WL8(-)z + LL	0.33	0.58	-5.88
0.9 DL + 1.6 WL1(+)x	0.00	0.69	0.02
0.9 DL + 1.6 WL2(+)x	0.00	-2.08	0.01
0.9 DL + 1.6 WL3(-)x	0.01	9.04	0.01
0.9 DL + 1.6 WL4(-)x	0.00	5.84	0.01
0.9 DL + 1.6 WL5(+)z	0.32	-0.60	5.90
0.9 DL + 1.6 WL6(+)z	0.34	-0.93	5.90
0.9 DL + 1.6 WL7(-)z	0.31	-2.39	-5.88
0.9 DL + 1.6 WL8(-)z	0.33	-2.72	-5.88

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DESIGN OF FOUNDATION MKD.F3

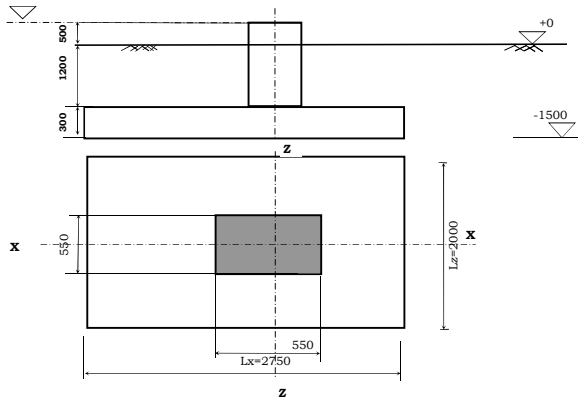
	About-X	About-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	2.75m	2.00m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

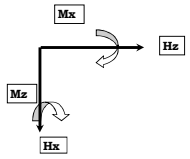
SYMBOLS AND NOTATIONS

Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Crane Load	CL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(2.75*2*0.3*2.5)+(0.55*0.55*1.7*2.5) =	5.41T
Weight of soil =	((2.75*2)-(0.55*0.55))*1.2*1.8 =	11.23T
U/S Base Pl	Total =	16.64T





Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = ######## 0.215m
Effective depth of slab along Z direction = ######## 0.195m
Clear projection along Z direction = 0.725m
Clear projection along X direction = 1.1m

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX	Hz (T)	MZ
DL+LL	0.00	6.15	0.02	0.0	0.00	22.79	0.00	0.03	0.02	0.01
DL+0.75*LL+0.75*WL1(+)x	0.00	4.23	0.02	0.0	0.00	20.87	0.00	0.03	0.02	0.01
DL+0.75*LL+0.75*WL2(+)x	0.00	2.93	0.02	0.0	0.00	19.57	0.00	0.03	0.02	0.01
DL+0.75*LL+0.75*WL3(-)x	0.01	8.15	0.02	0.0	0.00	24.79	0.01	0.03	0.02	0.01
DL+0.75*LL+0.75*WL4(-)x	0.00	6.65	0.02	0.0	0.00	23.29	0.00	0.03	0.02	0.01
DL+0.75*LL+0.75*WL5(+)z	0.15	3.63	2.78	0.0	0.00	20.27	0.15	5.55	2.78	0.31
DL+0.75*LL+0.75*WL6(+)z	0.16	3.48	2.78	0.0	0.00	20.12	0.16	5.55	2.78	0.32
DL+0.75*LL+0.75*WL7(-)z	0.15	2.79	-2.75	0.0	0.00	19.43	0.15	5.50	-2.75	0.29
DL+0.75*LL+0.75*WL8(-)z	0.16	2.64	-2.75	0.0	0.00	19.28	0.16	5.50	-2.75	0.31
0.6*DL+WL1(+)x	0.00	0.58	0.01	0.0	0.00	17.22	0.00	0.02	0.01	0.00
0.6*DL+WL2(+)x	0.00	-1.15	0.01	0.0	0.00	15.49	0.00	0.02	0.01	0.00
0.6*DL+WL3(-)x	0.00	5.80	0.01	0.0	0.00	22.44	0.00	0.02	0.01	0.01
0.6*DL+WL4(-)x	0.00	3.80	0.01	0.0	0.00	20.44	0.00	0.02	0.01	0.00
0.6*DL+WL5(+)z	0.20	-0.22	3.69	0.0	0.00	16.42	0.20	7.38	3.69	0.40
0.6*DL+WL6(+)z	0.21	-0.43	3.69	0.0	0.00	16.21	0.21	7.38	3.69	0.42
0.6*DL+WL7(-)z	0.19	-1.34	-3.68	0.0	0.00	15.30	0.19	7.35	-3.68	0.38
0.6*DL+WL8(-)z	0.20	-1.55	-3.68	0.0	0.00	15.09	0.20	7.35	-3.68	0.41

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+LL	22.79	0.00	0.01	0.02	0.03	2342.00	638.73	2025.51	552.41
DL+0.75*LL+0.75*WL1(+)x	20.87	0.00	0.01	0.02	0.03	2340.39	559.66	2024.12	484.03
DL+0.75*LL+0.75*WL2(+)x	19.57	0.00	0.01	0.02	0.03	3017.72	561.44	2609.92	485.57
DL+0.75*LL+0.75*WL3(-)x	24.79	0.01	0.01	0.02	0.03	2183.87	711.03	1888.75	614.94
DL+0.75*LL+0.75*WL4(-)x	23.29	0.00	0.01	0.02	0.03	2611.20	718.08	2258.34	621.04
DL+0.75*LL+0.75*WL5(+)z	20.27	0.15	0.31	2.78	5.55	61.43	3.38	53.13	2.92
DL+0.75*LL+0.75*WL6(+)z	20.12	0.16	0.32	2.78	5.55	57.57	3.35	49.79	2.90
DL+0.75*LL+0.75*WL7(-)z	19.43	0.15	0.29	-2.75	5.50	61.13	3.27	52.87	2.83
DL+0.75*LL+0.75*WL8(-)z	19.28	0.16	0.31	-2.75	5.50	57.15	3.24	49.43	2.81
0.6*DL+WL1(+)x	17.22	0.00	0.00	0.01	0.02	4425.10	758.59	3827.11	656.08
0.6*DL+WL2(+)x	15.49	0.00	0.00	0.01	0.02	23879.65	795.99	20652.67	688.42
0.6*DL+WL3(-)x	22.44	0.00	0.01	0.01	0.02	3145.49	1153.35	2720.42	997.49
0.6*DL+WL4(-)x	20.44	0.00	0.00	0.01	0.02	5252.59	1260.62	4542.78	1090.27
0.6*DL+WL5(+)z	16.42	0.20	0.40	3.69	7.38	38.01	2.06	32.87	1.78
0.6*DL+WL6(+)z	16.21	0.21	0.42	3.69	7.38	35.41	2.03	30.62	1.76
0.6*DL+WL7(-)z	15.30	0.19	0.38	-3.68	7.35	36.79	1.92	31.82	1.66
0.6*DL+WL8(-)z	15.09	0.20	0.41	-3.68	7.35	34.17	1.90	29.55	1.64

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ULTIMATE STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V	Mx	Mz
1.4 DL	0.00	5.69	0.02	0.0	0.00	14.241	0.04	0.01
1.4 DL + 1.7 LL	0.01	9.23	0.02	0.0	0.00	17.780	0.05	0.01
1.2 DL + 1.6 WL1(+)x + LL	0.01	3.99	0.02	0.0	0.00	12.539	0.04	0.01
1.2 DL + 1.6 WL2(+)x + LL	0.00	3.32	0.02	0.0	0.00	11.874	0.04	0.01
1.2 DL + 1.6 WL3(-)x + LL	0.01	12.34	0.02	0.0	0.00	20.893	0.04	0.02
1.2 DL + 1.6 WL4(-)x + LL	0.01	9.14	0.02	0.0	0.00	17.692	0.03	0.01
1.2 DL + 1.6 WL5(+)z + LL	0.32	2.70	5.91	0.0	0.00	11.255	11.82	0.64
1.2 DL + 1.6 WL6(+)z + LL	0.34	2.38	5.91	0.0	0.00	10.926	11.82	0.68
1.2 DL + 1.6 WL7(-)z + LL	0.31	0.91	-5.88	0.0	0.00	9.460	11.75	0.62
1.2 DL + 1.6 WL8(-)z + LL	0.33	0.58	-5.88	0.0	0.00	9.131	11.75	0.66
0.9 DL + 1.6 WL1(+)x	0.00	0.69	0.02	0.0	0.00	9.237	0.03	0.01
0.9 DL + 1.6 WL2(+)x	0.00	-2.08	0.01	0.0	0.00	6.465	0.03	0.00
0.9 DL + 1.6 WL3(-)x	0.01	9.04	0.01	0.0	0.00	17.592	0.03	0.01
0.9 DL + 1.6 WL4(-)x	0.00	5.84	0.01	0.0	0.00	14.390	0.02	0.01
0.9 DL + 1.6 WL5(+)z	0.32	-0.60	5.90	0.0	0.00	7.953	11.81	0.64
0.9 DL + 1.6 WL6(+)z	0.34	-0.93	5.90	0.0	0.00	7.625	11.81	0.68
0.9 DL + 1.6 WL7(-)z	0.31	-2.39	-5.88	0.0	0.00	6.158	11.77	0.62
0.9 DL + 1.6 WL8(-)z	0.33	-2.72	-5.88	0.0	0.00	5.829	11.77	0.65

CHECK FOR BEARING PRESSURE

Load Combination & Calculation of Base pressure

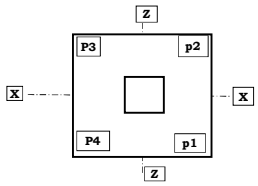
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Load Case	Design Load/moments			Base Pressure (T/m²)								
	V (T)	Mz (T-m)	Mx (T-m)	V/(Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/ Lx.(Lz)²	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
DL+LL	22.79	0.01	0.03	8.37	0.00	0.01	10.39	10.36	10.38	10.35	8.23	O.K
DL+0.75*LL+0.75*WL1(+)x	20.87	0.01	0.03	7.67	0.00	0.01	9.69	9.66	9.68	9.65	7.53	O.K
DL+0.75*LL+0.75*WL2(+)x	19.57	0.01	0.03	7.19	0.00	0.01	9.21	9.18	9.20	9.17	7.05	O.K
DL+0.75*LL+0.75*WL3(-)x	24.79	0.01	0.03	9.11	0.01	0.01	11.12	11.10	11.11	11.09	8.96	O.K
DL+0.75*LL+0.75*WL4(-)x	23.29	0.01	0.03	8.55	0.00	0.01	10.57	10.55	10.56	10.54	8.41	O.K
DL+0.75*LL+0.75*WL5(+)z	20.27	0.31	5.55	7.45	0.17	2.20	11.82	7.41	11.48	7.08	9.66	O.K
DL+0.75*LL+0.75*WL6(+)z	20.12	0.32	5.55	7.39	0.18	2.20	11.77	7.36	11.42	7.01	9.61	O.K
DL+0.75*LL+0.75*WL7(-)z	19.43	0.29	5.50	7.14	0.16	2.18	11.48	7.12	11.16	6.80	9.32	O.K
DL+0.75*LL+0.75*WL8(-)z	19.28	0.31	5.50	7.08	0.17	2.18	11.43	7.07	11.09	6.73	9.27	O.K
0.6*DL+WL1(+)x	17.22	0.00	0.02	6.33	0.00	0.01	8.34	8.32	8.33	8.32	6.18	O.K
0.6*DL+WL2(+)x	15.49	0.00	0.02	5.69	0.00	0.01	7.70	7.68	7.70	7.68	5.54	O.K
0.6*DL+WL3(-)x	22.44	0.01	0.02	8.24	0.00	0.01	10.25	10.24	10.25	10.23	8.09	O.K
0.6*DL+WL4(-)x	20.44	0.00	0.02	7.51	0.00	0.01	9.52	9.50	9.51	9.50	7.36	O.K
0.6*DL+WL5(+)z	16.42	0.40	7.38	6.03	0.22	2.93	11.18	5.32	10.74	4.89	9.02	O.K
0.6*DL+WL6(+)z	16.21	0.42	7.38	5.96	0.23	2.93	11.11	5.26	10.65	4.80	8.95	O.K
0.6*DL+WL7(-)z	15.30	0.38	7.35	5.62	0.21	2.92	10.75	4.91	10.33	4.49	8.59	O.K
0.6*DL+WL8(-)z	15.09	0.41	7.35	5.54	0.22	2.92	10.68	4.85	10.24	4.40	8.52	O.K

CALCULATION OF DESIGN PRESSURE

Load Combination & Calculation of Base pressure

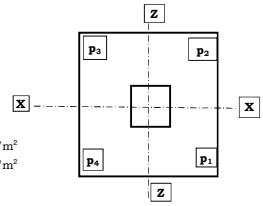
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Load Case	Design Load/moments			Base Pressure (T/m²)							
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/ Lx.(Lz)²	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)
1.4 DL	14.2	0.01	0.04	5.23	0.00	0.02	7.25	7.22	7.24	7.21	4.55
1.4 DL + 1.7 LL	17.8	0.01	0.05	6.53	0.01	0.02	8.56	8.52	8.54	8.50	5.86
1.2 DL + 1.6 WL1(+)x + LL	12.5	0.01	0.04	4.61	0.01	0.02	6.63	6.59	6.62	6.58	3.93
1.2 DL + 1.6 WL2(+)x + LL	11.9	0.01	0.04	4.36	0.00	0.01	6.38	6.35	6.37	6.34	3.68
1.2 DL + 1.6 WL3(-)x + LL	20.9	0.02	0.04	7.67	0.01	0.02	9.70	9.67	9.68	9.65	7.00
1.2 DL + 1.6 WL4(-)x + LL	17.7	0.01	0.03	6.50	0.01	0.01	8.52	8.49	8.51	8.48	5.82
1.2 DL + 1.6 WL5(+)z + LL	11.3	0.64	11.82	4.13	0.35	4.69	11.17	1.80	10.47	1.09	8.47
1.2 DL + 1.6 WL6(+)z + LL	10.9	0.68	11.82	4.01	0.37	4.69	11.07	1.70	10.33	0.95	8.37
1.2 DL + 1.6 WL7(-)z + LL	9.5	0.62	11.75	3.47	0.34	4.66	10.48	1.15	9.80	0.47	7.78
1.2 DL + 1.6 WL8(-)z + LL	9.1	0.66	11.75	3.35	0.36	4.66	10.38	1.05	9.66	0.33	7.68
0.9 DL + 1.6 WL1(+)x	9.2	0.01	0.03	3.39	0.00	0.01	5.41	5.38	5.40	5.38	2.71
0.9 DL + 1.6 WL2(+)x	6.5	0.00	0.03	2.37	0.00	0.01	4.39	4.36	4.39	4.36	1.69
0.9 DL + 1.6 WL3(-)x	17.6	0.01	0.03	6.46	0.01	0.01	8.48	8.46	8.47	8.45	5.78
0.9 DL + 1.6 WL4(-)x	14.4	0.01	0.02	5.29	0.00	0.01	7.30	7.28	7.29	7.27	4.60
0.9 DL + 1.6 WL5(+)z	8.0	0.64	11.81	2.92	0.35	4.68	9.95	0.59	9.26	-0.11	7.25
0.9 DL + 1.6 WL6(+)z	7.6	0.68	11.81	2.80	0.37	4.68	9.85	0.49	9.11	-0.25	7.15
0.9 DL + 1.6 WL7(-)z	6.2	0.62	11.77	2.26	0.34	4.67	9.27	-0.07	8.59	-0.74	6.57
0.9 DL + 1.6 WL8(-)z	5.8	0.65	11.77	2.14	0.36	4.67	9.17	-0.17	8.45	-0.88	6.47

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	29-05-2020	29-06-2020	26-10-2023	Date	29-05-2020	29-06-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM1z =Moment at the face of column 1z-1Z in T-m/m.

SF(oneway) = Oneway shear force at a distance 'd' from the face of column in T/m.

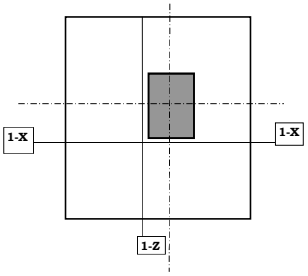
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2

= 2.98 m

Punching Area= ((0.55+2*0.5*0.195)*(0.55+2*0.5*0.195))

= 0.56 m^2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)

= 2.91 T/m^2

Area of footing, A_f= 5.50 m^2

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 41.84 T

Strength of Concrete in shear Vc1 = 124.28 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 132.28 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 114.59 T **Table 22.6.5.2_ (c)**

Design Vc= 114.59 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (41.84*10000)/(2980*195)

= 0.72 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$ ™	$S.F_{(oneway)}$ (T)	$S.F_{(punching)}$ (T)	V_u/b_0d	Punching check (Remarks)	Allowable shear ($0.75*\sqrt{(fc)}/6$ (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4 DL	4.55	2.75	4.03	1.20	2.41	22.48	0.39	OK	0.74	0.207	OK	0.60
1.4 DL + 1.7 LL	5.86	3.55	5.19	1.54	3.11	28.95	0.50	OK	0.74	0.266	OK	0.77
1.2 DL + 1.6 WL1(+)x + LL	3.93	2.38	3.48	1.03	2.08	19.41	0.33	OK	0.74	0.178	OK	0.51
1.2 DL + 1.6 WL2(+)x + LL	3.68	2.23	3.26	0.97	1.95	18.18	0.31	OK	0.74	0.167	OK	0.48
1.2 DL + 1.6 WL3(-)x + LL	7.00	4.24	6.20	1.84	3.71	34.58	0.60	OK	0.74	0.318	OK	0.92
1.2 DL + 1.6 WL4(-)x + LL	5.82	3.52	5.15	1.53	3.08	28.75	0.49	OK	0.74	0.264	OK	0.76
1.2 DL + 1.6 WL5(+)z + LL	8.47	5.12	7.50	2.23	4.49	41.84	0.72	OK	0.74	0.384	OK	1.11
1.2 DL + 1.6 WL6(+)z + LL	8.37	5.06	7.41	2.20	4.44	41.35	0.71	OK	0.74	0.38	OK	1.10
1.2 DL + 1.6 WL7(-)z + LL	7.78	4.71	6.89	2.04	4.12	38.43	0.66	OK	0.74	0.353	OK	1.02
1.2 DL + 1.6 WL8(-)z + LL	7.68	4.65	6.80	2.02	4.07	37.94	0.65	OK	0.74	0.349	OK	1.01
0.9 DL + 1.6 WL1(+)x	2.71	1.64	2.40	0.71	1.44	13.39	0.23	OK	0.74	0.123	OK	0.35
0.9 DL + 1.6 WL2(+)x	1.69	1.02	1.50	0.44	0.90	8.35	0.14	OK	0.74	0.077	OK	0.22
0.9 DL + 1.6 WL3(-)x	5.78	3.50	5.12	1.52	3.06	28.55	0.49	OK	0.74	0.262	OK	0.76
0.9 DL + 1.6 WL4(-)x	4.60	2.78	4.07	1.21	2.44	22.72	0.39	OK	0.74	0.209	OK	0.60
0.9 DL + 1.6 WL5(+)z	7.25	4.39	6.42	1.91	3.84	35.82	0.62	OK	0.74	0.329	OK	0.95
0.9 DL + 1.6 WL6(+)z	7.15	4.33	6.33	1.88	3.79	35.32	0.61	OK	0.74	0.325	OK	0.94
0.9 DL + 1.6 WL7(-)z	6.57	3.97	5.81	1.73	3.48	32.46	0.56	OK	0.74	0.298	OK	0.86
0.9 DL + 1.6 WL8(-)z	6.47	3.91	5.73	1.70	3.43	31.96	0.55	OK	0.74	0.294	OK	0.85

Load Case	kn (Ø=0.75)	ρ =	A_{st}
Ultimate Conditions			
1.4 DL	0.79	0.002	430
1.4 DL + 1.7 LL	1.02	0.002	430
1.2 DL + 1.6 WL1(+)x + LL	0.69	0.002	430
1.2 DL + 1.6 WL2(+)x + LL	0.64	0.002	430
1.2 DL + 1.6 WL3(-)x + LL	1.22	0.003	645
1.2 DL + 1.6 WL4(-)x + LL	1.02	0.002	430
1.2 DL + 1.6 WL5(+)z + LL	1.48	0.004	860
1.2 DL + 1.6 WL6(+)z + LL	1.46	0.004	860
1.2 DL + 1.6 WL7(-)z + LL	1.36	0.003	645
1.2 DL + 1.6 WL8(-)z + LL	1.34	0.003	645
0.9 DL + 1.6 WL1(+)x	0.47	0.001	215
0.9 DL + 1.6 WL2(+)x	0.29	0.001	215
0.9 DL + 1.6 WL3(-)x	1.01	0.002	430
0.9 DL + 1.6 WL4(-)x	0.8	0.002	430m
0.9 DL + 1.6 WL5(+)z	1.27	0.003	645
0.9 DL + 1.6 WL6(+)z	1.25	0.003	645
0.9 DL + 1.6 WL7(-)z	1.15	0.003	645
0.9 DL + 1.6 WL8(-)z	1.13	0.003	645

Location	Mux	ϕ Kn =	Kn	ρ=	Ast	Minm. Ast	Bar Dia	Spacing	Maxm	Spacing
At bottom	5.12	1.11	1.48	0.004	860.00	387.00	16ϕT	233.79	300	200
At top	0.00	0.00	0.00	0.0018	351.00	351.00	12ϕT	322.21	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **16ϕT** **200** c/c At Top **12ϕT** **200** c/c

Along Z Direction :

At Bottom **16ϕT** **200** c/c At Top **12ϕT** **200** c/c

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	29-05-2020	29-06-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
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Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008
[6] GENERAL ARRANGEMENT DRAWING FUEL UNLOADING STATION MD-216-49PR-EG-ST-GAD-2411

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Project Name	MANSOURAH MASSARAH GOLD PROJECT
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Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 1 FOUNDATION LOAD DATA

Global Project ID	900707
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Project Country	KINGDOM OF SAUDI ARABIA

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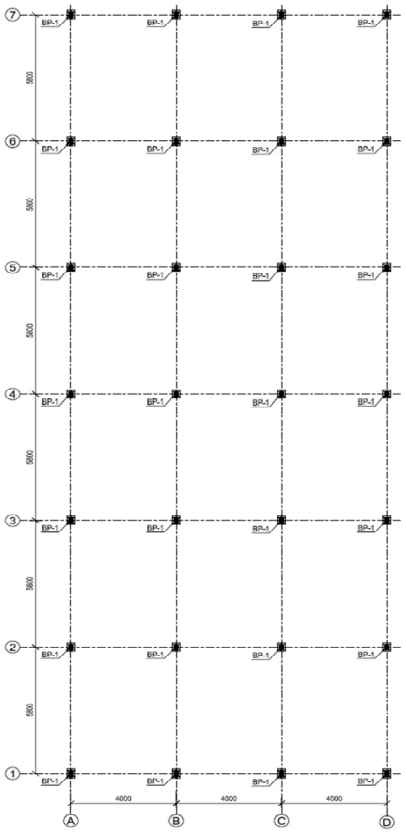
MMGP:POWER PLANT:Fuel unloading station shed

1 DL	Dead load			
2 LLROOF	Live load on roof			
21 WL1(+X)CASE-A	Wind load	Wind load in direction of +X Suction(from A grid to D grid)	side wind	**Sign convention**
22 WL2(+X)CASE-B	Wind load	Wind load in direction of +X Pressure(from A grid to D grid)	side wind	
23 WL3(-X)CASE-A	Wind load	Wind load in direction of -X Suction(from D grid to A grid)	side wind	FY +ve sign indicates vertical load downward
24 WL4(-X)CASE-B	Wind load	Wind load in direction of -X Pressure(from D grid to A grid)	side wind	-ve sign indicates vertical load upward
25 WL5(+Z)CASE-A	Wind load	Wind load in direction of +Z Suction(from line-1 to line-7)	gable wind	FX +ve sign indicates horizonal load in across direction from grid-A to D
26 WL6(+Z)CASE-B	Wind load	Wind load in direction of +Z Pressure(from line-1 to line-7)	gable wind	-ve sign indicates horizonal load in across direction from grid-D to A
27 WL7(-Z)CASE-A	Wind load	Wind load in direction of -Z Suction(from line-7 to line-1)	gable wind	FZ +ve sign indicates horizonal load in along direction from line 1 to 7
28 WL8(-Z)CASE-B	Wind load	Wind load in direction of -Z Pressure(from line-7 to line-1)	gable wind	-ve sign indicates horizonal load in along direction from line 7 to 1

Case-A and Case-B defined as per ASCE-7-16 fig-27.3-5

LOAD DATA TABLE

	Load cases	Horizontal Fx Ton	Vertical Fy Ton	Horizontal Fz Ton	
D	1 DL	0.0	1.8	0.0	Line-2,3,4,5,6
	2 LLROOF	0.0	1.8	0.0	
	21 WL1(+X)CASE-A	0.5	-0.7	0.0	
	22 WL2(+X)CASE-B	0.5	-2.2	0.0	
	23 WL3(-X)CASE-A	-0.5	2.1	0.0	
	24 WL4(-X)CASE-B	-0.5	0.5	0.0	
	25 WL5(+Z)CASE-A	0.0	-2.9	1.0	
	26 WL6(+Z)CASE-B	0.0	-3.1	1.0	
	27 WL7(-Z)CASE-A	0.0	-2.9	-1.0	
	28 WL8(-Z)CASE-B	0.0	-3.1	-1.0	
A	1 DL	0.0	1.8	0.0	
	2 LLROOF	0.0	1.8	0.0	
	21 WL1(+X)CASE-A	0.5	2.1	0.0	
	22 WL2(+X)CASE-B	0.5	0.5	0.0	
	23 WL3(-X)CASE-A	-0.5	-0.7	0.0	
	24 WL4(-X)CASE-B	-0.5	-2.2	0.0	
	25 WL5(+Z)CASE-A	0.0	-2.9	1.0	
	26 WL6(+Z)CASE-B	0.0	-3.1	1.0	
	27 WL7(-Z)CASE-A	0.0	-2.9	-1.0	
	28 WL8(-Z)CASE-B	0.0	-3.1	-1.0	
B	1 DL	0.0	4.0	0.0	
	2 LLROOF	0.0	3.4	0.0	
	21 WL1(+X)CASE-A	2.2	-1.1	0.0	
	22 WL2(+X)CASE-B	2.2	-4.0	0.0	
	23 WL3(-X)CASE-A	-2.2	3.6	0.0	
	24 WL4(-X)CASE-B	-2.2	0.6	0.0	
	25 WL5(+Z)CASE-A	-0.1	-5.4	0.8	
	26 WL6(+Z)CASE-B	-0.1	-5.8	0.8	
	27 WL7(-Z)CASE-A	-0.1	-5.4	-0.8	
	28 WL8(-Z)CASE-B	-0.1	-5.8	-0.8	
C	1 DL	0.0	4.0	0.0	
	2 LLROOF	0.0	3.6	0.0	
	21 WL1(+X)CASE-A	2.2	3.7	0.0	
	22 WL2(+X)CASE-B	2.2	0.6	0.0	
	23 WL3(-X)CASE-A	-2.2	-1.0	0.0	
	24 WL4(-X)CASE-B	-2.2	-4.0	0.0	
	25 WL5(+Z)CASE-A	-0.1	-5.7	0.8	
	26 WL6(+Z)CASE-B	-0.1	-6.0	0.8	
	27 WL7(-Z)CASE-A	-0.1	-5.7	-0.8	
	28 WL8(-Z)CASE-B	-0.1	-6.0	-0.8	
D	1 DL	0.0	2.0	0.0	Line-1,7
	2 LLROOF	0.0	0.9	0.0	
	21 WL1(+X)CASE-A	0.5	-0.4	0.0	
	22 WL2(+X)CASE-B	0.5	-1.2	0.0	
	23 WL3(-X)CASE-A	-0.5	1.1	0.0	
	24 WL4(-X)CASE-B	-0.5	-0.1	0.0	
	25 WL5(+Z)CASE-A	0.0	-0.2	2.2	
	26 WL6(+Z)CASE-B	0.0	-0.2	2.2	
	27 WL7(-Z)CASE-A	0.0	-2.7	-2.2	
	28 WL8(-Z)CASE-B	0.0	-2.8	-2.2	
A	1 DL	0.0	2.0	0.0	
	2 LLROOF	0.0	0.9	0.0	
	21 WL1(+X)CASE-A	0.5	1.1	0.0	
	22 WL2(+X)CASE-B	0.5	0.3	0.0	
	23 WL3(-X)CASE-A	-0.5	-0.4	0.0	
	24 WL4(-X)CASE-B	-0.5	-1.2	0.0	
	25 WL5(+Z)CASE-A	0.0	-0.2	2.2	
	26 WL6(+Z)CASE-B	0.0	-0.3	2.2	
	27 WL7(-Z)CASE-A	0.0	-2.8	-2.2	
	28 WL8(-Z)CASE-B	0.0	-2.8	-2.2	
B	1 DL	0.0	4.1	0.0	
	2 LLROOF	0.0	2.1	0.0	
	21 WL1(+X)CASE-A	0.0	-1.9	0.0	
	22 WL2(+X)CASE-B	0.0	-3.6	0.0	
	23 WL3(-X)CASE-A	0.0	3.4	0.0	
	24 WL4(-X)CASE-B	0.0	1.4	0.0	
	25 WL5(+Z)CASE-A	0.2	-2.7	3.7	
	26 WL6(+Z)CASE-B	0.2	-2.9	3.7	
	27 WL7(-Z)CASE-A	0.2	-3.8	-3.7	
	28 WL8(-Z)CASE-B	0.2	-4.0	-3.7	
C	1 DL	0.0	4.1	0.0	
	2 LLROOF	-0.1	1.5	0.0	
	21 WL1(+X)CASE-A	1.8	3.2	0.0	
	22 WL2(+X)CASE-B	1.9	1.8	0.0	
	23 WL3(-X)CASE-A	-1.9	-2.0	0.0	
	24 WL4(-X)CASE-B	-1.7	-3.8	0.0	
	25 WL5(+Z)CASE-A	0.2	-1.9	3.7	
	26 WL6(+Z)CASE-B	0.2	-2.1	3.7	
	27 WL7(-Z)CASE-A	0.2	-3.1	-3.7	
	28 WL8(-Z)CASE-B	0.2	-3.2	-3.7	



BASE PLATE LAYOUT(DIM IN MM)

All crane and wind load cases are exclusive.